EXP-012 — Hierarchical Document → Passage Retrieval

Production RAG v1 · evaluation-first · n = 20 questions / 22 evidence spans · scorers frozen, only topology changed

STATUS — THE INTERVENTION FAILED; THE DIAGNOSTIC SUCCEEDED

Hierarchical retrieval never beat the global control at any routing width. But the oracle-document diagnostic reached **0.950 / 19-of-20 with zero regressions and nothing absent at 300** — so global competition *is* causal, and the binding constraint is **document routing**, not passage scoring.

-0.050
A → D primary intervention
0 rescued, 1 regressed

0.950
oracle-document diagnostic
19/20, zero regressions

17/20
cases with every expected document routed into top 5

6.3%
of the corpus left after routing (900 of 14,209 chunks)

1. Why EXP-012 exists

AN-003 has shown the same shape since EXP-007: correct document at rank 2–6, answer chunk absent from the top 300. EXP-010 established that 21 of 22 answers are visible to the encoder; EXP-011 that query rewriting rescues zero cases. The remaining explanation was competition — the right passage ranked against ~14,209 chunks, nearly all from irrelevant documents.

2. Design, and the constraint that makes it interpretable

Stage 1 collapses each retriever's chunk ranking by document — a document ranks at its highest-ranked chunk and votes *once* — then fuses the two lists with RRF. Stage 2 restricts candidates to the routed documents and ranks them by their **full-corpus** scores.

BM25 term statistics are **not** recomputed inside the routed documents: the restriction sits in the scoring select only, so `n`, `avg_len` and `df` still come from the whole snapshot. Verified directly — a chunk's restricted score is bit-identical to its global score. Without this the experiment would have confounded topology with lexical re-weighting. Two planner tests confirm the GIN index survives the added predicate.

3. Reproduction gate

Cell A reproduced the control exactly — **0.775 / 15-of-20 / 17-of-22: PASS**.

4. Results

cell	macro R	full	spans@10	doc R	MRR	a@10	a@30	a@300
A — global raw hybrid (control)	0.775	15/20	17/22	0.925	0.449	5	4	2
B — BM25 routing → BM25 local	0.575	11/20	12/22	0.750	0.282	10	7	7
C — transformer routing → tx local	0.675	13/20	15/22	0.875	0.346	7	6	4
D — fused routing → fused local	0.725	14/20	16/22	0.875	0.421	6	4	4

cell	macro R	full	spans@10	doc R	MRR	a@10	a@30	a@300
ORACLE — golden doc → fused local	0.950	19/20	21/22	1.000	0.646	1	0	0

The oracle row is **ORACLE / DIAGNOSTIC / NOT DEPLOYABLE** — it reads the golden expected document and is excluded from every production metric. Tests assert it is not among the deployable configurations.

5. Document routing — the ceiling on everything

ranking	@1	@3	@5	@10
bm25	0.475	0.675	0.750	0.875
transformer	0.525	0.675	0.875	0.925
fused	0.525	0.800	0.875	0.950

At the preregistered `top_documents = 5`, every expected document is routed for only **17 of 20** cases — AN-001, AN-012, OA-004 each lose one before Stage 2 begins. **Stage-1 ceiling 0.850**: even a perfect passage ranker could not exceed it. D reached 14/20, so it does not even reach its own ceiling.

6. Why hierarchy lost

- **Routing drops documents.** Three cases lose an expected document outright — an error of *exclusion* the global system structurally cannot make. OA-004 is the regression.
- **Complementarity degrades.** The hierarchical fusion bonus is **+1 case** (best component 13 → fused 14), against **+4** globally. The third time this project has seen a change erode the disagreement RRF feeds on.
- **Rerankable headroom shrinks** — see §8.

Local pools average 900 chunks (6.34% of the corpus), so competition really was removed. It simply did not help.

> **THE ORACLE: PASSAGE RANKING IS NEARLY FINE**
>
> With perfect routing, the *same* retrievers, scores and fusion reach **0.950 / 19-of-20**, document recall **1.000**, MRR **0.646**, and **nothing absent at 300**. Rescued: AN-001, AN-006, AN-011, AN-012 — with **zero regressions**.
>
> Global competition was genuinely suppressing answer-bearing passages. What fails is our ability to *route*, not our ability to rank passages once routed.

7. AN-003 — the exception, now precisely characterised

stage	value
fused document rank	**3 — routed correctly**
chunks in the top-5 pool	3,125
global evidence rank	**absent@300**
fused hierarchical evidence rank	**absent@300**
chunks in its own document	141

stage	value
oracle evidence rank	**29**

Its document is routed correctly, yet hierarchy does not help — the top-5 pool still holds 3,125 chunks, 22% of the corpus. Only when competition is cut to its own 141-chunk document does the evidence appear at all, and even then at **rank 29**. **AN-003 is the one case the oracle does not rescue** — a genuine within-document passage-ranking failure, and now a single precisely-characterised defect rather than a property of the system.

8. Failure taxonomy and reranker ceilings

case	classification
AN-001	DOCUMENT ROUTING FAILURE
AN-003	**WITHIN DOCUMENT PASSAGE RANKING FAILURE**
AN-006	MIXED OR UNCLEAR
AN-011	MIXED OR UNCLEAR
AN-012	DOCUMENT ROUTING FAILURE

cell	1–10	11–30	31–50	51–100	101–300	abs	c@30	c@50	c@100
A global	17	1	1	1	0	2	0.818	0.864	0.909
D hierarchical	16	2	0	0	0	4	0.818	0.818	0.818
ORACLE (diagnostic)	21	1	0	0	0	0	1.000	1.000	1.000

Hierarchy **reduced** rerankable headroom: the perfect-reranker ceiling falls from 0.909 (A, pool 100) to 0.818 (D, flat). With perfect routing it is **1.000 at a pool of just 30**.

9. Exploratory sensitivity — routing width

routing width	D macro recall	D full	all docs routed	mean local pool
N = 3	0.750	15/20	16/20	500
N = 5 (primary)	0.725	14/20	17/20	900
N = 10	0.775	15/20	19/20	1701

Run only after the primary result was frozen and **labelled exploratory**. Hierarchy **converges to the global control** as routing widens: at N = 10 it matches A exactly. It never exceeds it at any width. The N=3 vs N=5 difference is one case —

noise at this sample size. The clearest statement of the result: **the best hierarchical system is the one that stops being hierarchical.**

10. Limitations

- n = 20 / 22 spans; one case is 5 percentage points. No significance claims.
- The oracle is an upper bound with a perfect router. It says the headroom exists, not that it is reachable.
- One collapse rule was tested (highest-ranked chunk, one vote per document). Score aggregation, top-k voting or document-level embeddings might route better and are untested.
- AN-003's oracle rank of 29 is a single case.
- **EXP-NULL remains BLOCKED** — still no measured no-retrieval floor.

11. Was global competition causal?

Yes — and that is the surprise. Every previous mechanism this project proposed turned out not to be causal: chunk size (three times), enrichment, encoder visibility, query formulation. This one is: removing cross-document competition entirely takes the system from 0.775 to 0.950 with zero regressions.

But **the intervention still failed**, because achievable routing cannot realise it. Every document the router drops is a case lost outright. The honest summary: *global competition is a real bottleneck; document → passage retrieval as built is not the way to remove it.*

12. What the measurements justify next

1. **Work on document routing, not passage ranking.** The oracle says passage ranking is already worth 19/20; routing recall@5 of 0.875 is the constraint — a genuinely different problem from anything tried so far.
2. **Do not deploy hierarchy at any N.** It never beats global and adds an exclusion failure mode global retrieval cannot have.
3. **A reranker is justified for the first time — but only behind good routing.** With perfect routing a perfect reranker over 30 candidates reaches 1.000; behind current routing the ceiling is 0.818, *worse* than global's 0.909. Sequence matters.
4. **AN-003 deserves a failure report**, not another system-wide experiment.
5. **Keep measuring the fusion bonus.** Three changes have now improved components while degrading the ensemble.

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